

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2015
CE 2107
(Construction Technology and Materials)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.



Section – A

1. (a) What is Building stone? Describe any four tests by which the quality of stones can be checked. (12)
- (b) What are the harmful constituents of brick clay? Write down the functions of alumina, silica, lime and iron oxide of brick earth. (12)
- (c) How can we determine the quality of bricks in the field? (06)
- (d) What is the standard size of bricks in our country? Do you think that the existing size is economical? Justify your answer. (05)

2. (a) Write down the differences between: (09)
 - (i) Initial setting time and Final setting time of cement.
 - (ii) False setting and Flash setting of cement.
 - (iii) Hydration and soundness of cement.
- (b) Draw a neat flow diagram of manufacturing Portland cement by wet process. (05)
- (c) What is "Bulking of Sand"? Draw the qualitative gradation curve for well graded, gap graded and uniformly graded sample. (05)
- (d) There are three different varieties of sand available in a construction site. Sieve analysis data of sand type 'A' is given in the following table. The fineness modules (FM) of sand type 'B' and 'C' are 2.20 and 2.40, respectively. To prepare a concrete mix, 20 kg sand of FM = 2.80 is required. What are the possible ways to get the required sand for the cement mix? Calculate the amount of each varieties of sand for the required 20 kg sand sample. (16)

Sieve No.	Weight Retained (gm)
No. 4	10
No. 8	60
No. 16	90
No. 20	110
No. 30	130
No. 50	70
No. 100	25
No. 200	10
Pan	5

3. (a) What is "Bleeding" and "Segregation" of concrete? What are their effects on quality and strength of concrete? (10)
- (b) What are the factors affecting sufficient workability? Describe the slump test for measuring workability. (12)
- (c) What are the assumptions behind concrete mix design by ACI method? (04)
- (d) Write short notes on – (09)
 - (i) Creep of concrete
 - (ii) Chemical attack of concrete
 - (iii) Admixtures in concrete.

4. (a) Define the following terms: (06)
 (i) Available water holding capacity (ii) Cation-exchange capacity (iii) Pedology
- (b) What is meant by seasoning of timber? What are the advantages of seasoning of timber? (07)
- (c) Write down the uses of mild steel, cast iron, wrought iron, aluminum and plastics. (15)
- (d) Explain briefly the quality of good bricks. (07)



Section – B

5. (a) What is foundation? What are the factors that dictate the choice of a foundation? (08)
- (b) Draw the neat sketch of the following foundations. Also state why and when they are best suited for – i) Strip foundation ii) Pile foundation iii) Raft foundation iv) Pad foundation. (12)
- (c) Draw pressure bulb for pile foundation. (05)
- (d) Write short notes on i) pre cast and ii) in-situ concrete pile. (05)
- (e) What are the main features that make difference between concrete shell structure and conventional RCC structure? (05)
6. (a) What do you mean by load bearing masonry and infill masonry? Show with neat sketches the different types of masonry bonds. (10)
- (b) What are the main disadvantages of cavity walls? How can you overcome those shortcomings? (07)
- (c) What are arches and lintel? Classify lintel according to their functionality. (08)
- (d) What is composite flooring system? Describe the aspects for using composite structures. (10)
7. (a) What is the difference between floor and roof? What are the major types of flooring system? (05)
- (b) What is waffle slab? When are they used? (05)
- (c) Draw neat sketches of the following windows – i) Bay window ii) Bow window iii) Fixed frame window iv) Sliding/Gliding sash window. (08)
- (d) Draw a schematic diagram for a typical septic tank for a single household. (07)
- (e) Give short notes with necessary figures of the following types of bridge – (10)
 i) Arch bridge ii) Cable stayed bridge iii) Truss bridge iv) Girder bridge v) Suspension bridge
8. (a) What are the different types of road pavements? Draw the cross section of a typical rigid pavement with different layers. (06)
- (b) Mention the comparative merits and demerits of English bond and Flemish bond. (06)
- (c) Show with neat sketches the different types of masonry joint painting. (08)
- (d) What is the main difference between shoring and shielding? Explain with neat sketch. (08)
- (e) Mention the most common types of trenching hazards and trenching collapse. (07)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2015.
Hum 2117
(Sociology for Planners)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.



Section – A

1. (a) What is sociology? (05)
(b) Describe the importance of studying sociology for the students of Urban and Regional Planning. (15)
(c) What is society? (05)
(d) Explain different types of societies with distinctive characteristics. (10)
2. (a) Differentiate between Association and Institution. (10)
(b) Why are human societies stratified? Justify the reasons with example. (10)
(c) Explain social structure and its elements. (15)
3. (a) Differentiate between subculture and counter culture. (10)
(b) Explain the elements or carriers of culture. (10)
(c) What is cultural lag? (05)
(d) What are the causes and consequences of cultural lag? (10)
4. (a) What is industrialization? (05)
(b) How can you measure degree or level of urbanization? (10)
(c) Describe the impact of industrialization on third world countries like Bangladesh. (10)
(d) Describe the major functions of modern family. (10)

Section – B

5. (a) What is migration? (05)
(b) What are the causes of migration? (10)
(c) Explain the relationship between urbanization and migration. (10)
(d) Discuss the impact of rural-urban migration in Bangladesh. (10)
 6. (a) Define leadership. (05)
(b) Describe the major types of leadership. (15)
(c) Discuss the roles of leadership in rural Bangladesh with examples. (15)
 7. (a) What is urban ecology? (05)
(b) Discuss the concentric zone model of urban ecology. (15)
(c) Explain the level of ecological explanation. (15)
 8. (a) Narrate with example the impact of population on environment. (15)
(b) Explain the impact of globalization in 3rd world countries like Bangladesh. (20)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2015.
URP 2115
(Urban Planning *Techniques*)

Full Marks: 210

Time: 3.0 hrs

- N.B.: i) Answer any **three** questions from each section in separate script
ii) Figures in the right margin indicate marks
iii) Assume data if necessary

<u>Section - A</u>		Marks
1. (a)	What is your understanding about Urban Development Plan? "Urban Development Plan is a vital planning instrument for controlling dynamic urban areas like our country"-justify the statement with proper example.	(15)
(b)	Write about the issues on which Urban Development Plan works for ensuring a balanced development.	(10)
(c)	Write about the common forms of Urban Development Plan found in Bangladesh.	(10)
2. (a)	Master Plan is not usually practiced because of some difficulties in practical field-specify these issues with necessary example.	(06)
(b)	Make a comparative analysis among Strategic Plan, Structure Plan and Detailed Area Plan practiced in our country.	(15)
(c)	How will you define Local Plan? Write about the common forms of Local Plan. Which one of these common forms is practiced in our country?	(09)
(d)	What is the concept of Bankable Projects in Detailed Area Plan? How priority of projects is determined here?	(05)
3. (a)	Specify some pre-considerations of Participatory Area Planning techniques.	(06)
(b)	Suppose, you are working as a Town Planner in Khulna Development Authority (KDA). Your authority is going to develop a residential complex in a fringe area of Khulna city-which technique will you prefer between Guided Land Development and Land Consolidation for area development? Justify your answer.	(13)
(c)	Write about the implementation challenges of Land Banking and Land Ceiling technique for Bangladesh.	(08)
(d)	Make a comparison between Infrastructure Led Development Plan and Area Improvement Plan techniques of Participatory Area Planning.	(08)
4. (a)	Write about the directions regarding necessary road arrangement and parking provision mentioned in "Dhaka Metropolitan Building Construction Rule, 2007".	(15)
(b)	Write short notes on any four:	(20)
i.	Setback Rule	
ii.	Maximum Ground Coverage	
iii.	Floor Area Ratio	
iv.	Land Ceiling	
v.	Fire Drill	

Section – B

5. (a) What are the data required for Planning? Also provide the criteria for any one data. (15)
(b) Provide a brief description on the park, recreation and open space need assessment technique. (10)
(c) Show the difference between the Gompertz Curve and the Logistic Curve. (10)
 6. (a) The current amount of space used by commercial businesses and offices in 2010 is 8640000 sq. ft. and the population is 8640. Total expected number of additional people by the year 2020 is 864, estimate the need for new commercial space. Assume that the need for new space per person will increase at the same rate throughout your planning time period. (15)
(b) Describe the steps of Suitability Analysis with an appropriate example. (20)
 7. (a) "Demolition vs Renewal"- Which do you think is the correct choice and why? (10)
(b) Provide a brief overview on the Build-Out Analysis methods. (15)
(c) Show the levels of intervention in Urban Renewal with the help of a diagram. (10)
 8. (a) Provide a conceptual framework for Housing Choice. (10)
(b) With the help of a single example, show the difference between Base and Income Multiplier. (05)
(c) Briefly describe the Household Projection Process. (10)
(d) What are the problems and assumptions of the Locational Quotients (LQ) approach? (10)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2015

URP-2113
(Site and Area Planning)

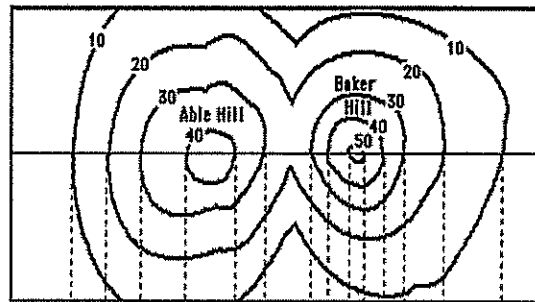
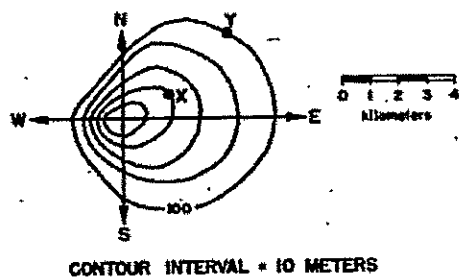
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Provide appropriate diagrams to portray Kevin Lynch's site planning process. (10)
- (b) Give the various elements of a Site Inventory Checklist. (15) (10)
- (c) Draw the elevation of the following sections: (10)



- (d) d) Find the void ratio with the following data: $V_A=5\text{m}^3$, $V_W=3000000\text{cm}^3$ and $V_S=10\text{m}^3$. (05)
2. (a) What are the scopes of site selection process? Show the benefits of selecting a suitable site with the help of an example. (15)
- (b) Briefly describe the criteria for the evaluation of the existing vegetation. (08)
- (c) Show the types of erosion with the help of appropriate diagrams (05)
- (d) What are the factors affecting the rate of infiltration in a site? (07)
3. (a) Provide the steps for RPOS planning processes. (10)
- (b) What are the key RPOS planning challenges? (05)
- (c) Briefly describe the different climate elements on the following attributes (provide appropriate diagrams wherever necessary): (15)
 - i. Representation or measurement
 - ii. Control or modification.
- (d) When will it be require to prepare a micro-climate map? (05)
4. (a) "Ecological sustainability can be achieved through land use controls" – Justify the statement. (10)
- (b) With the help of examples, show the need for land use controls? (10)
- (c) A rectangular plot of land is divided into 4 zones namely A, B, C and D. The following is the composition of each zone: (15)

A= Residential with building height/story 2, 3, 4 and 5.

B= Commercial with building height/story 4, 6, 8 and 12.

C= Industrial with building height/story 5, 7 and 9.

D= A mixture of each of the above.

$A \Rightarrow B$ (75%), $B \Rightarrow C$ (25%), $C \Rightarrow D$ (50%), $D \Rightarrow B$ (75%) and $B \Rightarrow A$ (75%)

Show the above zonal map and also the land use change map for the above.

Section – B

5. Write down the objectives, performance criteria and prescriptive (5x7) measures for **any five** of the following elements of residential (35) subdivision.
- Density
 - Allotment layout
 - Allotment design
 - Setbacks
 - Solar access
 - Private open space
 - Landscaping
6. (a) What do you understand by neighbourhood planning? State the (10) purposes of neighbourhood planning.
- (b) "Site Planning is related to *spirit of the place, character of the place, nature of the project, and behavioral studies*" – justify the statement with your rationalities by providing pertinent examples, if necessary. (10)
- (c) Write down any five good features of a planned neighborhood. (05)
- (d) "The size of the neighborhood unit is expressed as the size of the (10) population and the extent of the geographical area" – briefly describe the statement with proper examples.
7. (a) Chalk out three comparative critical examples (good design and bad (10) design) of subdivision planning and neighborhood design.
- (b) Which pattern of subdivision plan is more suitable than others in case of (10) Bangladesh especially in the flat region as per your understanding and why? State your rationalities.
- (c) Annotate **any three** of the following pattern of subdivision. (5x3) (15)
- Gridiron Pattern
 - Modified grid pattern
 - Curvilinear pattern
 - Rectilinear pattern
 - Cul-de-sac pattern
8. (a) What are the regulatory bases of Private Residential Land Development (05) Rules 2012?
- (b) Briefly describe the available technologies for site and area planning? (10)
- (c) Write down the general conditions for the approval of a private (20) residential housing project under the Private Residential Land Development Rules 2012.

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 2nd Year 1st Term Examination, 2015.

URP 2191

(Statistics for Planners - I)

Full Marks: 210

Time: 3.0 hrs

N.B.: i) Answer any **three** questions from each section in separate script

ii) Necessary graphs

iii) Table: Normal (t) table, Z-table, χ^2 - table



Section - A

Marks

1. (a) Discuss briefly about the importance of statistics for planners. (08)

(b) Draw (i) Histogram (ii) Ogive (iii) Frequency Polygon from the following data:- (09)

Class	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	4	7	10	15	8	3

(c) If x_1, x_2, x_3 and x_4 are non-zero positive integer, obtain the relation between A.M, G.M and H.M. (10)

(d) What is standard deviation? Calculated variance from following data:- (08)

Marks obtained	5-10	10-15	15-20	20-25	25-30
Standards	11	10	15	8	3

2. (a) Given below the frequency distribution of the marks obtained by 90 students. Compute (i) Arithmetic mean (ii) Median (iii) Mean deviation about mode. (15)

(b) What do you mean by co-efficient of variation, explain? (05)

(c) If $\bar{x}$ and σ denote respectively mean and standard deviation of a set of n non-negative observations, then prove that $\bar{x}\sqrt{1-n} \geq \sigma$ (15)

3. (a) Explain the central moment and raw moments. What are the difference between them? (10)

(b) Calculate (i) Third quantile Q_3 (ii) 7th decile D_7 (iii) 67th percentile P_{67} from the following data: 5, 2, 1, 0, -6, 7, -4 (15)

(c) The workers age are given below: (10)

Age (year): 22, 25, 29, 41, 44, 49

(i) Compute Kurtosis

(ii) Comment the nature of the distribution

4. (a) Define Skewness and Kurtosis. Compute Karl Pearson's Co-efficient of Skewness from the following data:- (12)

20-25	25-30	30-35	35-40	40-45	45-50
12	8	20	15	8	5

(b) What are difference between Skew curve and normal curve? (06)

(c) According to statistical analysis, in which subject the student has performed best and worst respectively? (17)

Subject	Earned grade	Average	Standard Deviation
A	90	95	1
B	80	75	2
C	70	40	5
D	70	40	10

Section – B

Marks

5. (a) Define with example (any four): (16)
 (i) Sample space (ii) Event (iii) Dependent and independent event
 (iv) Mutually exclusive event (v) Conditional probability.
- (b) Four components are inspected and three events A, B, C are defined (19)
 as follows:
 A = All four components are found to be in proper working order.
 B = At least two components are found to be defective.
 C = Exactly one components is found defective.
- (i) How many elementary events exist in the sample space?
 (ii) Define above events by the element of set.
 (iii) Interpret the following events: $B \cup C$, $B \cap C$, $A \cup C$.
6. (a) Explain discrete probability distribution and expectation of a random variable. (08)
- (b) In a region there are 8 buildings, 5 hut and 7 tin-shed building. If 4 houses are (12)
 selected randomly, determine the probability that:
 (i) All four are tin-shed building
 (ii) At least one is tin-shed building
- (c) Is the following functions are probability distribution function, why? If exist, find $E(x)$ of (15)
 each problem.
- (i)

x	2	1	-1
g(x)	212	k	3k

 (ii)

x	0	1	2	3
h(x)	0.5	0.4	0.3	4
- (iii)
$$\int(x) = \frac{-3e^x}{10}; x \geq 0$$

$$= x; x < 0$$
7. (a) Define Binomial Distribution with mentioning necessary assumptions. Hence, define (13)
 the mean of the Binomial distribution.
- (b) In a city there are 15,000 families. The number of each family is 10. The probability of (14)
 each number has job is 0.40. How many families are you expected to have:
 (i) At least one member has job.
 (ii) Exactly two member have job.
 (iii) None of them has job.
- (c) Write down three properties of each of the following distributions: (08)
 (i) Binomial distribution (ii) Poisson distribution (iii) Normal distribution.
8. (a) In a city, the average height of building is 50m and Standard Deviation is 10m. Find (17)
 the probability of:
 (i) Standard height building i.e. 20m to 60m.
 (ii) Low height building i.e. <20m.
 (iii) High rise building i.e. $\geq 70m$.
 (iv) In there are 10,000 buildings, estimate the number of the high rise buildings in
 that city.
- (b) Fit the following data in Poisson distribution, then test the goodness of fit with 5% level (18)
 of significance.

Defective product in a packet	0	1	2	3	4	5
Number of packets count	5	7	8	6	3	1

